Chapter Eight: Results and Analysis

This chapter aims to present what the system actually produced in practice. Accordingly, it includes the results of applying UPPMS (screenshots, reports, statistics, and charts that illustrate system performance), followed by a comparative analysis with existing systems in terms of speed, accuracy, cost, ease of use, and features.

Blank boxes marked **[SCREENSHOT PLACEHOLDER]** and cells marked *[ ]* are left empty intentionally for the author to fill with real captures, measured values, or external references when available.

8-1 System Application Results

This section presents the outcomes of running UPPMS on the Syrian Private University demo environment. The results are organized as screenshots of the main flows, summary reports/statistics, and charts of system performance.

8-1-1 Screenshots of Main System Flows

The following figures illustrate the principal end-to-end flows of UPPMS. Each figure should be replaced with an actual screen capture from the running system.

[SCREENSHOT PLACEHOLDER] Insert screenshot here:	**User Registration / Login**	Show student/supervisor registration or login screen (XML-matched account preferred).

Figure 8-1. User Registration / Login

[SCREENSHOT PLACEHOLDER] Insert screenshot here:	**University Admin Dashboard**	Show admin home / university workspace after login.

Figure 8-2. University Admin Dashboard

[SCREENSHOT PLACEHOLDER] Insert screenshot here:	**XML Authorized-Users Import**	Show Users → XML Import with an uploaded file and import result summary.

Figure 8-3. XML Authorized-Users Import

[SCREENSHOT PLACEHOLDER] Insert screenshot here:	**Project Proposal Submission (Student)**	Show proposal form and list of submitted proposals.

Figure 8-4. Project Proposal Submission (Student)

[SCREENSHOT PLACEHOLDER] Insert screenshot here:	**Supervisor Proposal Review / Approval**	Show supervisor view of a pending proposal with approve/reject actions.

Figure 8-5. Supervisor Proposal Review / Approval

[SCREENSHOT PLACEHOLDER] Insert screenshot here:	**Academic Track / Student Progress Timeline**	Show track stages (locked / in progress / passed) for a demo student.

Figure 8-6. Academic Track / Student Progress Timeline

[SCREENSHOT PLACEHOLDER] Insert screenshot here:	**AI Project Ideation**	Show AI idea suggestions screen (Gemini) if used in the demo.

Figure 8-7. AI Project Ideation

[SCREENSHOT PLACEHOLDER] Insert screenshot here:	**Project Workspace / Tasks**	Show an approved project with tasks (optionally AI-generated).

Figure 8-8. Project Workspace / Tasks

[SCREENSHOT PLACEHOLDER]

Insert screenshot here:

Defense Committee Management

Show committee list/members for SPU demo committees.

Figure 8-9. Defense Committee Management

[SCREENSHOT PLACEHOLDER]

Insert screenshot here:

Genetic Scheduling Dashboard

Show schedule generation candidates and/or an approved defense schedule.

Figure 8-10. Genetic Scheduling Dashboard

8-1-2 Reports and Statistics

Table 8-1 summarizes quantitative results observed from the demo deployment. Values left blank should be filled after a measured run (seeded demo counts may be used where already known).

Table 8-1. System Application Statistics (Demo / Measured)

Metric	Value	Notes
University under test	syrian private uni (SPU)	Demo tenant
Active students (demo)	28	SpuCampusDemoSeeder
Active supervisors (demo)	10	SpuCampusDemoSeeder
Active projects (demo)	28	Approved proposals
Defense committees (demo)	4	Seminar / technical / final
Available rooms (demo)	7	Including premium rooms
Automated test cases in repository	141	50 Unit + 91 Feature
Average API response time (common CRUD)	[ ]	Measure and fill
Schedule generation time (GA, demo load)	[ ]	Measure and fill
Proposal approval turnaround (demo observation)	[ ]	Optional
User-satisfaction score (survey)	[ ]	If a survey was conducted

Table 8-2 is reserved for any exported/system-generated reports (PDF exports, schedule reports, progress summaries). Attach the report file or paste a screenshot of the report view.

Table 8-2. Generated Reports Produced by the System

Report	Produced?	Where shown / file name
Approved defense schedule view	[ ]	[ ]
Student progress / track summary	[ ]	[ ]
XML import comparison summary	[ ]	[ ]
Other: [ ]	[ ]	[ ]

8-1-3 Charts Illustrating System Performance

The following chart placeholders should be replaced with actual charts (e.g., Excel, Python, or dashboard exports) once measurements are available.

[SCREENSHOT PLACEHOLDER]

Insert screenshot here:

Performance Chart — Response Time / Load

Replace with a chart of API latency or scheduling runtime vs. input size.

Figure 8-11. Performance Chart — Response Time / Load

[SCREENSHOT PLACEHOLDER]

Insert screenshot here:

Usage / Outcome Chart

Replace with a chart of proposals, approvals, defenses, or progress statuses.

Figure 8-12. Usage / Outcome Chart

A short textual analysis of the charts should be written here after the figures are inserted:

[Analysis of Figure 8-11 and Figure 8-12 — to be completed by the author.]

8-2 Comparative Analysis with Existing Systems

This section compares UPPMS with existing approaches/systems used for managing graduation projects (manual tools, spreadsheets, generic PMS, or institutional systems). Criteria follow the guide: speed, accuracy, cost, ease of use, and features.

Fill the competitor columns with the real systems or approaches you compared against (names + references). Leave cells blank if a fair measurement is not available.

Table 8-3. Comparative Analysis: UPPMS vs Existing Systems

Criterion	UPPMS	Existing System / Approach A *[Name / Ref]*	Existing System / Approach B *[Name / Ref]*
Speed (scheduling / coordination)	Automated genetic scheduling + admin approval	*[ ]*	*[ ]*
Accuracy (conflicts, prerequisites, isolation)	Hard/soft constraints, track prerequisites, tenant isolation, XML matching	*[ ]*	*[ ]*
Cost	*[ ]* (hosting / licenses — fill)	*[ ]*	*[ ]*
Ease of use	Role-based SPA (Admin / Supervisor / Student), bilingual UI	*[ ]*	*[ ]*
Core features	Multi-tenant PMS, XML registration, proposals, AI ideation/tasks, tracks, committees, GA scheduling	*[ ]*	*[ ]*
Multi-university isolation	Yes (TenantScope)	*[ ]*	*[ ]*
Academic track enforcement	Yes	*[ ]*	*[ ]*
AI assistance	Yes (Gemini ideation + task breakdown)	*[ ]*	*[ ]*

Discussion of the comparison (to be completed after filling Table 8-3):

[Comparative discussion — strengths of UPPMS, limitations, and where existing systems remain preferable.]

References used for the comparison (if any):

[Reference 1]

[Reference 2]

[Reference 3]